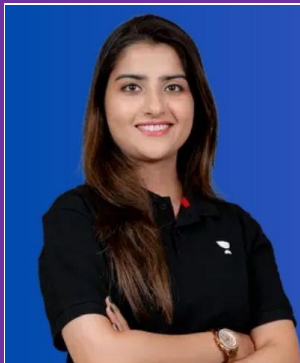




Seep Pahuja
Biology Educator
(NEET UG)



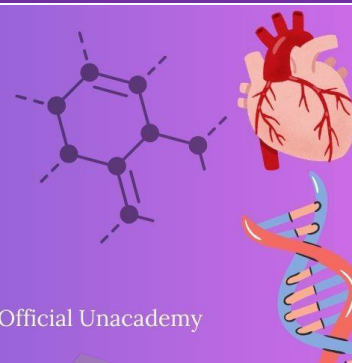
seep.pahuja



Seep Pahuja Biology - Official Unacademy

LAST 10 YEARS

NEET PYQ



Chapter 3 : Plant Kingdom

1. Which one of the following is the characteristic feature of gymnosperms? [NEET-2025]

- (1) Seeds are enclosed in fruits
- (2) Seeds are naked
- (3) Seeds are absent
- (4) Gymnosperms have flowers for reproduction

2. Match List-I with List-II. [NEET-2025]

List-I

List-II

- | | |
|-----------------|------------------|
| A. Pteridophyte | I. Salvia |
| B. Bryophyte | II. Ginkgo |
| C. Angiosperm | III. Polytrichum |
| D. Gymnosperm | IV. Salvinia |

Choose the option with all correct matches.

- (1) A-III, B-IV, C-II, D-I
- (2) A-IV, B-III, C-I, D-II
- (3) A-III, B-IV, C-I, D-II
- (4) A-IV, B-III, C-II, D-I

3. Given below are the stages in the life cycle of pteridophytes. Arrange the following stages in the correct sequence.

- A. Prothallus stage
- B. Meiosis in spore mother cells
- C. Fertilisation
- D. Formation of archegonia and antheridia in gametophyte.
- E. Transfer of antherozoids to the archegonia in presence of water.

Choose the correct answer from the options given below:

- | | |
|-------------------|-------------------|
| (1) B, A, D, E, C | (2) B, A, E, C, D |
| (3) D, E, C, A, B | (4) E, D, C, B, A |

4. In bryophytes, the gemmae help in which one of the following? [NEET-2025]

- (1) Sexual reproduction
- (2) Asexual reproduction
- (3) Nutrient absorption
- (4) Gaseous exchange

5. The correct sequence of events in the life cycle of bryophytes is [NEET-2025]

- A. Fusion of antherozoid with egg.
- B. Attachment of gametophyte to substratum.
- C. Reduction division to produce haploid spores.
- D. Formation of sporophyte.
- E. Release of antherozoids into water.

Choose the correct answer from the options given below :

- (1) D, E, A, C, B
- (2) B, E, A, C, D
- (3) B, E, A, D, C
- (4) D, E, A, B, C

6. Which one of the following is not found in Gymnosperms? [Re-NEET-2024]

- (1) Sieve cells
- (2) Albuminous cells
- (3) Tracheids
- (4) Vessels

7. Identify the incorrect pair : [Re-NEET-2024]

- (1) Sphenopsida – Adiantum
- (2) Pteropsida – Dryopteris
- (3) Psilopsida – Psilotum
- (4) Lycopsida – Selaginella

8. Which of the following is the correct match? [Re-NEET-2024]

- (1) Gymnosperms : Cedrus, Pinus, Sequoia
- (2) Angiosperms : Wolffia, Eucalyptus, Sequoia
- (3) Bryophytes : Polytrichum, Polysiphonia, Sphagnum
- (4) Pteridophytes : Equisetum, Ginkgo, Adiantum

9. Which classes of algae possess pigment fucoxanthin and pigment phycoerythrin, respectively?

- (1) Chlorophyceae and Rhodophyceae [NEET-2023-MANIPUR]
- (2) Rhodophyceae and Phaeophyceae
- (3) Phaeophyceae and Chlorophyceae
- (4) Phaeophyceae and Rhodophyceae

10. Match List-I with List-II.

[NEET-2023-MANIPUR]

List-I	List-II
(A) Pteropsida	(I) Psilotum
(B) Lycopsidea	(II) Equisetum
(C) Psilopsida	(III) Adiantum
(D) Sphenopsida	(IV) Selaginella

Choose the correct answer from the options given below :

- (1) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
- (2) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
- (3) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
- (4) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

11. Given below are two statements : One labelled as Assertion A and the other labelled as Reason R:

Assertion A : The first stage of gametophyte in the life cycle of moss is protonema stage. [NEET-2023]

Reason R : Protonema develops directly from spores produced in capsule.

In the light of the above statements, choose the most appropriate answer from options given below:

- (1) Both A and R are correct and R is the correct explanation of A
- (2) Both A and R are correct but R is NOT the correct explanation of A
- (3) A is correct but R is not correct
- (4) A is not correct but R is correct

12. Identify the pair of heterosporous pteridophytes among the following :

[NEET-2023]

- (1) Lycopodium and Selaginella
- (2) Selaginella and Salvinia
- (3) Psilotum and Salvinia
- (4) Equisetum and Salvinia

13. Given below are two statements :

[NEET-2023]

Assertion A : In gymnosperms the pollen grains are released from the microsporangium and carried by air currents.

Reason R : Air currents carry the pollen grains to the mouth of the archegonia where the male gametes are discharged and pollen tube is not formed.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both A and R are true and R is the correct explanation of A
- (2) Both A and R are true but R is NOT the current explanation of A
- (3) A is true but R is false
- (4) A is false but R is true

14. Hydrocolloid carrageen is obtained from :

[NEET-2022]

- (1) Phaeophyceae and Rhodophyceae
- (2) Rhodophyceae only
- (3) Phaeophyceae only
- (4) Chlorophyceae and Phaeophyceae

15. Which of the following is incorrectly matched ?

[NEET-2022]

- (1) Ulothrix – Mannitol
- (2) Porphyra - Floridian Starch
- (3) Volvox – Starch
- (4) Ectocarpus – Fucoxanthin

16. Match the plant with the kind of life cycle it exhibits:

[NEET-2022]

List-I

List-II

- | | |
|---------------|---|
| (a) Spirogyra | (i) Dominant diploid sporophyte vascular plant, with highly reduced male or female gametophyte |
| (b) Fern | (ii) Dominant haploid free living gametophyte |
| (c) Funaria | (iii) Dominant diploid sporophyte alternating with reduced gametophyte called prothallus |
| (d) Cycas | (iv) Dominant haploid leafy gametophyte alternating with partially dependent multicellular sporophyte |

Choose the correct answer from the options given below:

- (1) (a)–(ii), (b)–(iii), (c)–(iv), (d)–(i)
- (2) (a)–(iii), (b)–(iv), (c)–(i), (d)–(ii)
- (3) (a)–(ii), (b)–(iv), (c)–(i), (d)–(iii)
- (4) (a)–(iv), (b)–(i), (c)–(ii), (d)–(iii)

17. Match List - I with List - II :

[Re-NEET-2022]

List - I

List - II

- | | |
|-------------------|-------------------|
| (a) Chlamydomonas | (i) Moss |
| (b) Cycas | (ii) Pteridophyte |
| (c) Selaginella | (iii) Alga |
| (d) Sphagnum | (iv) Gymnosperm |

Choose the correct answer from the options given below :

- (1) (a)–(iii), (b)–(i), (c) - (ii), (d) - (iv)
- (2) (a)–(iii), (b)–(iv), (c) - (ii), (d) - (i)
- (3) (a)–(iii), (b)–(ii), (c) - (i), (d) - (iv)
- (4) (a)–(ii), (b)–(iii), (c) - (i), (d) - (iv)

18. Read the following statements and identify the characters related to the alga shown in the diagram:

[Re-NEET-2022]



- (a) It is a member of Chlorophyceae
- (b) Food is stored in the form of starch
- (c) It is a monoecious plant showing oogonium and antheridium
- (d) Food is stored in the form of laminarin or mannitol
- (e) It shows dominance of pigments chlorophyll a, c and Fucoxanthin.

Choose the correct answer from the options given below :

- (1) (a) and (b) only
- (2) (a), (b) and (c) only
- (3) (a), (c) and (d) only
- (4) (c), (d) and (e) only

19. Gemmae are present in :

[NEET-2021]

- (1) Mosses
- (2) Pteridophytes
- (3) Some Gymnosperms
- (4) Some Liverworts

20. Genera like *Selaginella* and *Salvinia* produce two kinds of spores. Such plants are known as:

[NEET-2021]

- (1) Homosorus
- (2) Heterosorus
- (3) Homosporous
- (4) Heterosporous

21. Which of the following plants is monoecious?

[NEET-2021]

- (1) *Carica papaya*
- (2) *Chara*
- (3) *Marchantia polymorpha*
- (4) *Cycas circinalis*

22. Which of the following algae contains mannitol as reserve food material ?

[NEET-2021]

- (1) *Ectocarpus*
- (2) *Gracilaria*
- (3) *Volvox*
- (4) *Ulothrix*

23. Which of the following algae produce Carrageen ? [NEET-2021]
- (1) Green algae
 - (2) Brown algae
 - (3) Red algae
 - (4) Blue-green algae
24. Floridean starch has structure similar to: [NEET-2020]
- (1) Laminarin and cellulose
 - (2) Starch and cellulose
 - (3) Amylopectin and glycogen
 - (4) Mannitol and align
25. Which of the following pairs is of unicellular algae? [NEET-2020]
- (1) Chlorella and Spirulina
 - (2) Laminaria and Sargassum
 - (3) Gelidium and Gracilaria
 - (4) Anabaena and Volvox
26. Strobili or cones are found in : [NEET-2020]
- (1) Equisetum
 - (2) Salvinia
 - (3) Pteris
 - (4) Marchantia
27. Phycoerythrin is the major pigment in : [NEET-2020 COVID-19]
- (1) Red algae
 - (2) Blue green algae
 - (3) Green algae
 - (4) Brown algae
28. Which of the following statements is incorrect about gymnosperms ? [NEET-2020 COVID-19]
- (1) They are heterosporous
 - (2) Male and female gametophytes are free living
 - (3) Most of them have narrow leaves with thick cuticle
 - (4) Their seeds are not covered
29. Male and female gametophytes do not have an independent free living existence in :- [NEET-2020 COVID-19]
- (1) Pteridophytes
 - (2) Algae
 - (3) Angiosperms
 - (4) Bryophytes

30. Phloem in gymnosperms lacks : [NEET-2019]
- (1) Albuminous cells and sieve cells
 - (2) Sieve tubes only
 - (3) Companion cells only
 - (4) Both sieve tubes and companion cells
31. From evolutionary point of view, retention of the female gametophyte with developing young embryo on the parent sporophyte for some time, is first observed in : [NEET-2019]
- (1) Liverworts
 - (2) Mosses
 - (3) Pteridophytes
 - (4) Gymnosperms
32. Which of the following statement is correct? [NEET-2018]
- (1) Ovules are not enclosed by ovary wall in gymnosperms
 - (2) Selaginella is heterosporous, while Salvinia is homosporous
 - (3) Horsetails are gymnosperms
 - (4) Stems are usually unbranched in both Cycas and Cedrus
33. Which one is wrongly matched ? [NEET-2018]
- (1) Uniflagellate gametes - Polysiphonia
 - (2) Biflagellate zoospores - Brown algae
 - (3) Gemma cups - Marchantia
 - (4) Unicellular organism - Chlorella
34. Winged pollen grains are present in [NEET-2018]
- (1) Mustard
 - (2) Cycas
 - (3) Mango
 - (4) Pinus
35. An example of colonial alga is : [NEET-2017]
- (1) Volvox
 - (2) Ulothrix
 - (3) Spirogyra
 - (4) Chlorella
36. Zygotic meiosis is characteristic of; [NEET-2017]
- | | |
|-------------------|----------------|
| (1) Fucus | (2) Funaria |
| (3) Chlamydomonas | (4) Marchantia |

37. Life cycle of Ectocarpus and Fucus respectively are: [NEET-2017]

- (1) Diplontic, Haplodiplontic
- (2) Haplodiplontic, Diplontic
- (3) Haplodiplontic, Haplontic
- (4) Haplontic, Diplontic

38. Select the mismatch [NEET-2017]

- (1) Cycas – Dioecious
- (2) Salvinia – Heterosporous
- (3) Equisetum – Homosporous
- (4) Pinus – Dioecious

39. Select the correct statement :- [NEET-I 2016]

- (1) Gymnosperms are both homosporous and heterosporous
- (2) Salvinia, Ginkgo and Pinus all are gymnosperms
- (3) Sequoia is one of the tallest trees
- (4) The leaves of gymnosperms are not well adapted to extremes of climate

40. In bryophytes and pteridophytes, transport of male gametes requires :- [NEET-I 2016]

- (1) Wind
- (2) Insects
- (3) Birds
- (4) Water

41. Which of the following would appear as the pioneer organisms on bare rocks? [NEET-I 2016]

- (1) Lichens
- (2) Liverworts
- (3) Mosses
- (4) Green algae

42. The ovule of an angiosperm is technically equivalent to : [NEET-II 2016]

- (1) megaspore mother cell
- (2) megaspore
- (3) megasporangium
- (4) megasporophyll

43. Conifers are adapted to tolerate extreme environmental conditions because of :

[NEET-II 2016]

- (1) thick cuticle
- (2) presence of vessels
- (3) broad hardy leaves
- (4) superficial stomata

44. Which one of the following statements is wrong?

[NEET-II 2016]

- (1) Agar-agar is obtained from Gelidium and Gracilaria
- (2) Laminaria and Sargassum are used as food
- (3) Algae increase the level of dissolved oxygen in the immediate environment
- (4) Algin is obtained from red algae, and carrageenan from brown algae.

ANSWER KEY

1. (2)
2. (2)
3. (1)
4. (2)
5. (3)
6. (4)
7. (1)
8. (1)
9. (4)
10. (2)
11. (1)
12. (2)
13. (3)
14. (2)
15. (1)
16. (1)
17. (2)
18. (2)
19. (4)
20. (4)
21. (2)
22. (1)
23. (3)
24. (3)
25. (1)
26. (1)
27. (1)
28. (2)
29. (3)
30. (4)
31. (3)
32. (1)
33. (1)
34. (4)
35. (1)
36. (3)
37. (2)
38. (4)
39. (3)
40. (4)
41. (1)
42. (3)
43. (1)
44. (4)